

CSCE 3530 Final Network Project Report

Small Office Network Built in Proxmox

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This report documents the design, configuration, and testing of a small office-style network using virtual machines in Proxmox. The project includes a router/firewall VM, an internal server VM, and two Ubuntu Desktop client VMs. The network was tested for routing, NAT, DHCP, DNS, HTTP, SSH, and general connectivity.

Project Overview

The goal of this project was to create a functional small office network in a virtual lab environment. Proxmox was used as the virtualization platform. The network separates the outside/home network from the internal project network using a Router-Firewall-VM. The internal Server-VM provides DHCP, DNS, and Apache web services to two Ubuntu Desktop client machines.

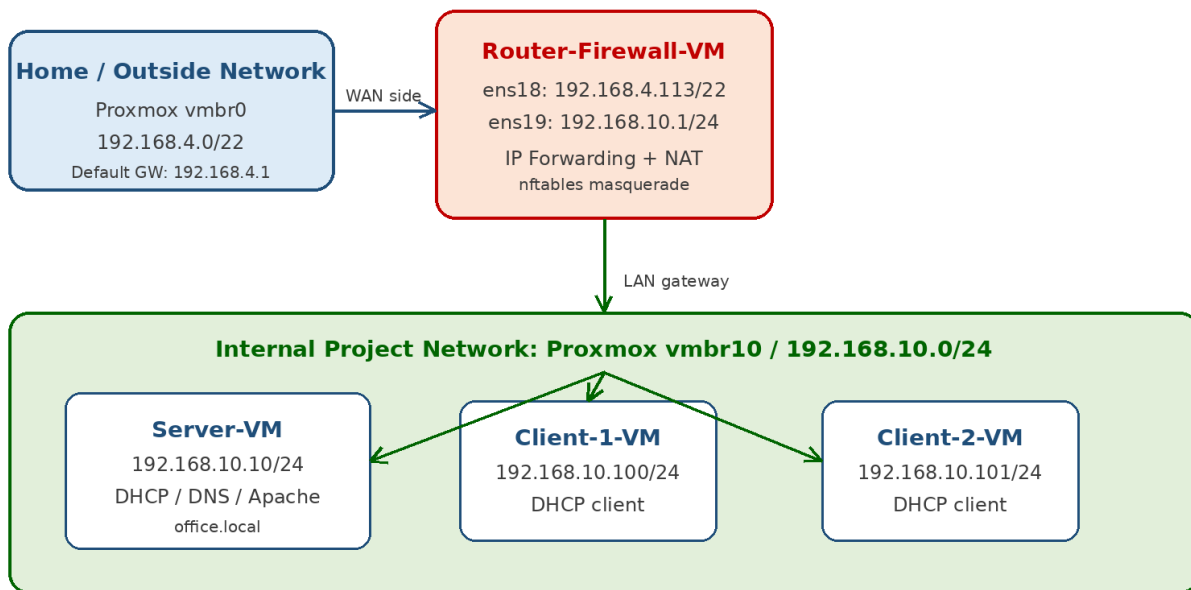
Objectives of the Project

- Build a virtualized office network using Proxmox.
- Configure a router/firewall VM with two interfaces, routing, and NAT.
- Configure a server VM to provide DHCP, DNS, and HTTP web services.
- Configure two client VMs to receive addressing automatically through DHCP.
- Verify connectivity with ping, DNS lookup, HTTP/curl testing, and traceroute.
- Document the final design with screenshots and explanations.

Network Design Explanation

The network uses two Proxmox bridges. The outside bridge, vmbro, connects the router VM to the home network. The internal bridge, vmbro10, connects the router VM, server VM, and client VMs together on the private 192.168.10.0/24 subnet. The router acts as the default gateway for the internal subnet and performs NAT so internal clients can reach the internet.

Small Office Network Topology



Clients receive DHCP from Server-VM, use Server-VM for DNS, and reach the internet through Router-Firewall-VM.

Figure 1. Network topology diagram showing the router, server, clients, subnet, and IP addressing.

The Proxmox inventory confirms the project was implemented using separate virtual machines for the Router-Firewall-VM, Server-VM, Client-1-VM, and Client-2-VM. Keeping each network role in its own VM helped simulate a small office network with a dedicated router/firewall, internal server, and client systems.

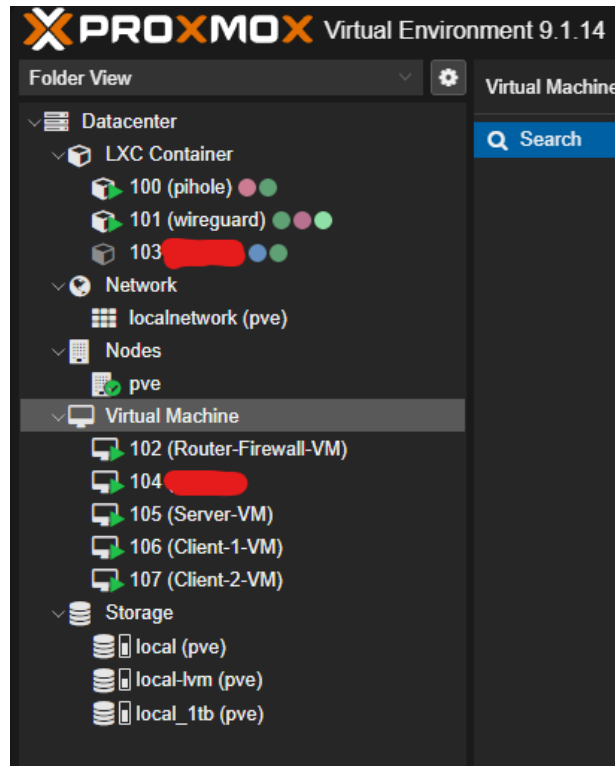


Figure 2. Proxmox VM inventory showing the project virtual machines used for the network lab.

Device	Interface	IP Address	Purpose
Router-Firewall-VM	ens18	192.168.4.113/22	Outside/home network on vmbro
Router-Firewall-VM	ens19	192.168.10.1/24	Internal gateway on vmbro10
Server-VM	ens18	192.168.10.10/24	DHCP, DNS, Apache web server
Client-1-VM	ens18	192.168.10.100/24	DHCP client
Client-2-VM	ens18	192.168.10.101/24	DHCP client

Description of Networking Protocols Used

Protocol / Service	Use in This Project
DHCP	The Server-VM assigns automatic IP addresses, gateway, DNS, and domain information to the client VMs.
DNS	BIND9 on the Server-VM resolves internal names such as <code>www.office.local</code> and <code>router.office.local</code> .
HTTP	Apache on the Server-VM hosts an internal web page available at <code>http://www.office.local</code> .
NAT	nftables masquerading on the Router-Firewall-VM allows internal clients to reach outside networks.
ICMP	Ping was used to confirm connectivity between clients, server, router, and the internet.
Traceroute	Traceroute confirmed that client traffic exits the internal network through the router at 192.168.10.1.
SSH	SSH was enabled so each VM could be managed from the Windows terminal through the router jump host.

Device Configurations

Router-Firewall-VM

The Router-Firewall-VM has two virtual network adapters. ens18 connects to vmbr0 and received an outside network address of 192.168.4.113/22. ens19 connects to vmbr10 and uses the internal gateway address 192.168.10.1/24. The router was configured with IPv4 forwarding and nftables NAT/forwarding rules.

```
yaboy9k@router-firewall-vm:~$ ip -br a
lo                UNKNOWN    127.0.0.1/8 ::1/128
ens18             UP          192.168.4.113/22 metric 100 fd0b:c76b:40cf:b70a:be24:11ff:fece:5a5b/64 fe80::be24:11ff:fece:5a5b/64
ens19             UP          192.168.10.1/24 fe80::be24:11ff:fe76:59a4/64
yaboy9k@router-firewall-vm:~$ ip route
default via 192.168.4.1 dev ens18 proto dhcp src 192.168.4.113 metric 100
192.168.4.0/22 dev ens18 proto kernel scope link src 192.168.4.113 metric 100
192.168.4.1 dev ens18 proto dhcp scope link src 192.168.4.113 metric 100
192.168.4.96 dev ens18 proto dhcp scope link src 192.168.4.113 metric 100
192.168.10.0/24 dev ens19 proto kernel scope link src 192.168.10.1
yaboy9k@router-firewall-vm:~$ _
```

Figure 3. Router-Firewall-VM interface and route configuration.

```
yaboy9k@router-firewall-vm:~$ sudo nano /etc/nftables.conf
yaboy9k@router-firewall-vm:~$ sudo nft -f /etc/nftables.conf
yaboy9k@router-firewall-vm:~$ sudo nft list ruleset
table ip nat {
    chain postrouting {
        type nat hook postrouting priority srcnat; policy accept;
        oifname "ens18" ip saddr 192.168.10.0/24 masquerade
    }
}
table inet filter {
    chain forward {
        type filter hook forward priority filter; policy drop;
        iifname "ens19" oifname "ens18" accept
        iifname "ens18" oifname "ens19" ct state established,related accept
    }
}
```

Figure 4. nftables NAT and forwarding rules on the Router-Firewall-VM.

Important router commands used:

```
ip -br a
ip route
sysctl net.ipv4.ip_forward
sudo nft list ruleset
```

Server-VM

The Server-VM was configured with a static address of 192.168.10.10/24, default gateway 192.168.10.1, and internal services for DHCP, DNS, and Apache HTTP. The server successfully reached both the router and the internet through the Router-Firewall-VM.

```

yaboy9k@server-vm:~$
yaboy9k@server-vm:~$ ip -br a
ip route
ping -c 4 192.168.10.1
ping -c 4 8.8.8.8
ping -c 4 google.com
lo                UNKNOWN          127.0.0.1/8 ::1/128
ens18             UP                192.168.10.10/24 fe80::be24:11ff:feef:130c/64
default via 192.168.10.1 dev ens18 proto static
192.168.10.0/24 dev ens18 proto kernel scope link src 192.168.10.10
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data.
64 bytes from 192.168.10.1: icmp_seq=1 ttl=64 time=0.281 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=64 time=0.296 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=64 time=0.362 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=64 time=0.410 ms

--- 192.168.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3060ms
rtt min/avg/max/mdev = 0.281/0.337/0.410/0.051 ms
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=2.79 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=3.19 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=116 time=3.10 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=116 time=3.64 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 2.785/3.181/3.644/0.307 ms
PING google.com (142.250.217.110) 56(84) bytes of data.
64 bytes from sea09s30-in-f14.1e100.net (142.250.217.110): icmp_seq=1 ttl=111 time=23.0 ms
64 bytes from sea09s30-in-f14.1e100.net (142.250.217.110): icmp_seq=2 ttl=111 time=23.6 ms
64 bytes from sea09s30-in-f14.1e100.net (142.250.217.110): icmp_seq=3 ttl=111 time=23.1 ms
64 bytes from sea09s30-in-f14.1e100.net (142.250.217.110): icmp_seq=4 ttl=111 time=23.0 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 22.969/23.175/23.634/0.268 ms
yaboy9k@server-vm:~$ |

```

Figure 5. Server-VM IP configuration, default route, and connectivity tests.

DHCP Configuration

ISC DHCP Server was configured to listen on ens18 and provide addresses in the 192.168.10.100-192.168.10.200 range. It also provides the default gateway 192.168.10.1 and DNS server 192.168.10.10 to clients.

```

host.
yaboy9k@server-vm:~$ sudo nano /etc/default/isc-dhcp-server
yaboy9k@server-vm:~$ sudo nano /etc/dhcp/dhcpd.conf
yaboy9k@server-vm:~$ sudo systemctl restart isc-dhcp-server
yaboy9k@server-vm:~$ ^C
yaboy9k@server-vm:~$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; e>
   Active: active (running) since Tue 2026-07-07 18:49:02 UTC; 13s ago
  Invocation: a7d7f4cf82e24415970f3c5415a246e0
     Docs: man:dhcpd(8)
    Main PID: 17008 (dhcpd)
       Tasks: 1 (limit: 2093)
      Memory: 4M (peak: 4.1M)
         CPU: 24ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─17008 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp>

Jul 07 18:49:02 server-vm sh[17008]: Wrote 0 leases to leases file.
Jul 07 18:49:02 server-vm dhcpd[17008]: PID file: /run/dhcp-server/dhcp>
Jul 07 18:49:02 server-vm dhcpd[17008]: Wrote 0 leases to leases file.
Jul 07 18:49:02 server-vm dhcpd[17008]: Listening on LPF/ens18/bc:24:11>
Jul 07 18:49:02 server-vm sh[17008]: Listening on LPF/ens18/bc:24:11:ef>
Jul 07 18:49:02 server-vm sh[17008]: Sending on LPF/ens18/bc:24:11:ef>
Jul 07 18:49:02 server-vm sh[17008]: Sending on Socket/fallback/fallb>
Jul 07 18:49:02 server-vm dhcpd[17008]: Sending on LPF/ens18/bc:24:11>
Jul 07 18:49:02 server-vm dhcpd[17008]: Sending on Socket/fallback/fa>
Jul 07 18:49:02 server-vm dhcpd[17008]: Server starting service.
lines 1-22/22 (END)

```

Figure 6. ISC DHCP Server service running on the Server-VM.

DHCP configuration summary:

- Subnet: 192.168.10.0/24
- DHCP range: 192.168.10.100 - 192.168.10.200
- Router/default gateway: 192.168.10.1
- DNS server: 192.168.10.10
- Domain: office.local

DNS Configuration

BIND9 was configured as the internal DNS server for the office.local domain. The zone includes records for www.office.local, server.office.local, and router.office.local.

```

yaboy9k@server-vm:~$ sudo named-checkconf
yaboy9k@server-vm:~$ sudo named-checkzone office.local /etc/bind/db.office.local
zone office.local/IN: loaded serial 2026070701
OK
yaboy9k@server-vm:~$ sudo systemctl restart bind9
yaboy9k@server-vm:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; pri>
   Active: active (running) since Tue 2026-07-07 18:52:15 UTC; 5s ago
 Invocation: 7baf779af92445448f84dcc5473a6862
    Docs: man:named(8)
   Main PID: 17117 (named)
    Status: "running"
     Tasks: 6 (limit: 2093)
    Memory: 5.9M (peak: 5.9M)
       CPU: 45ms
    CGroup: /system.slice/named.service
            └─17117 /usr/sbin/named -f -u bind

Jul 07 18:52:15 server-vm named[17117]: DNS format error from 198.97.19>
Jul 07 18:52:15 server-vm named[17117]: FORMERR resolving './NS/IN': 19>
Jul 07 18:52:15 server-vm named[17117]: DNS format error from 199.7.83.>
Jul 07 18:52:15 server-vm named[17117]: FORMERR resolving './NS/IN': 19>
Jul 07 18:52:15 server-vm named[17117]: DNS format error from 199.7.91.>
Jul 07 18:52:15 server-vm named[17117]: FORMERR resolving './NS/IN': 19>
Jul 07 18:52:15 server-vm named[17117]: DNS format error from 192.58.12>
Jul 07 18:52:15 server-vm named[17117]: FORMERR resolving './NS/IN': 19>
Jul 07 18:52:15 server-vm named[17117]: network unreachable resolving '>
Jul 07 18:52:15 server-vm named[17117]: resolver priming query complete>
lines 1-23/23 (END)

```

Figure 7. BIND9 DNS service running and zone file check passing.

```

yaboy9k@server-vm:~$ nslookup www.office.local 192.168.10.10
Server:          192.168.10.10
Address:         192.168.10.10#53

Name:   www.office.local
Address: 192.168.10.10

yaboy9k@server-vm:~$ nslookup router.office.local 192.168.10.10
Server:          192.168.10.10
Address:         192.168.10.10#53

Name:   router.office.local
Address: 192.168.10.1

```

Figure 8. DNS lookups resolving www.office.local and router.office.local using the Server-VM.

HTTP / Apache Configuration

Apache was installed on the Server-VM and used to host a simple internal web page. The page is reachable by IP address and by the DNS name `www.office.local` after DNS was configured.

```
yaboy9k@server-vm:~$ sudo nano /etc/systemd/resolved.conf
yaboy9k@server-vm:~$ sudo systemctl restart systemd-resolved
yaboy9k@server-vm:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub
    DNS Servers: 192.168.10.10 1.1.1.1 8.8.8.8
    DNS Domain: office.local

Link 2 (ens18)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS
                DNSSEC=no/unsupported
    DNS Servers: 1.1.1.1 8.8.8.8
    Default Route: yes
yaboy9k@server-vm:~$ nslookup www.office.local
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   www.office.local
Address: 192.168.10.10

yaboy9k@server-vm:~$ curl http://www.office.local
<!DOCTYPE html>
<html>
<head>
  <title>CSCE 3530 Network Project</title>
</head>
<body>
  <h1>CSCE 3530 Small Office Network</h1>
  <p>This page is hosted on the internal Server-VM at 192.168.10.10.</p>
  <p>Clients can access this page using the DNS name www.office.local.</p>
</body>
```

Figure 9. Server-VM resolver configuration and successful HTTP test using `www.office.local`.

DHCP Lease Verification

The DHCP lease file verifies that the Server-VM issued addresses to both client systems. Client-1-VM received 192.168.10.100 and Client-2-VM received 192.168.10.101.

```

yaboy9k@server-vm:~$ sudo cat /var/lib/dhcp/dhcpd.leases | tail -n 80
[sudo: authenticate] Password:
# The format of this file is documented in the dhcpd.leases(5) manual page.
# This lease file was written by isc-dhcp-4.4.3-P1

# authoring-byte-order entry is generated, DO NOT DELETE
authoring-byte-order little-endian;

lease 192.168.10.100 {
    starts 3 2026/07/08 01:58:42;
    ends 3 2026/07/08 02:08:42;
    cltt 3 2026/07/08 01:58:42;
    binding state active;
    next binding state free;
    rewind binding state free;
    hardware ethernet bc:24:11:65:cb:a6;
    uid "\001\274$\021e\313\246";
    client-hostname "yaboy9k-Standard-PC-i440FX-PIIX-1996";
}
lease 192.168.10.101 {
    starts 3 2026/07/08 02:03:22;
    ends 3 2026/07/08 02:13:22;
    cltt 3 2026/07/08 02:03:22;
    binding state active;
    next binding state free;
    rewind binding state free;
    hardware ethernet bc:24:11:58:a1:94;
    uid "\001\274$\021X\241\224";
    client-hostname "yaboy9k1-Standard-PC-i440FX-PIIX-1996";
}
lease 192.168.10.100 {
    starts 3 2026/07/08 02:03:42;
    ends 3 2026/07/08 02:13:42;
    cltt 3 2026/07/08 02:03:42;
    binding state active;
    next binding state free;
    rewind binding state free;
    hardware ethernet bc:24:11:65:cb:a6;
    uid "\001\274$\021e\313\246";
    client-hostname "yaboy9k-Standard-PC-i440FX-PIIX-1996";
}

```

Figure 10. DHCP lease file showing active leases for Client-1-VM and Client-2-VM.

Network Testing and Results

Client-1-VM Testing

Client-1-VM received its IP address automatically through DHCP. It used 192.168.10.1 as the default gateway and 192.168.10.10 as the DNS server. The client was able to ping the router, ping the server, reach the internet, resolve `www.office.local`, access the web page, and traceroute through the router.

```

yaboy9k@yaboy9k-Standard-PC-i440FX-PIIX-1996: ~
yaboy9k@yaboy9k-Standard-PC-i440FX-PIIX-1996:~$ ip -br a
lo                UNKNOWN          127.0.0.1/8 ::1/128
ens18             UP                192.168.10.100/24 fe80::be24:11ff:fe65:cba6/64
yaboy9k@yaboy9k-Standard-PC-i440FX-PIIX-1996:~$ ip route
default via 192.168.10.1 dev ens18 proto dhcp src 192.168.10.100 metric 100
192.168.10.0/24 dev ens18 proto kernel scope link src 192.168.10.100 metric 100
yaboy9k@yaboy9k-Standard-PC-i440FX-PIIX-1996:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (ens18)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 192.168.10.10
    DNS Servers: 192.168.10.10
    DNS Domain: office.local
    Default Route: yes
yaboy9k@yaboy9k-Standard-PC-i440FX-PIIX-1996:~$ ping -c 4 192.168.10.1
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data.
64 bytes from 192.168.10.1: icmp_seq=1 ttl=64 time=0.481 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=64 time=0.572 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=64 time=0.533 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=64 time=0.649 ms

--- 192.168.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3099ms
rtt min/avg/max/mdev = 0.481/0.558/0.649/0.061 ms
yaboy9k@yaboy9k-Standard-PC-i440FX-PIIX-1996:~$ ping -c 4 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=64 time=0.598 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=64 time=0.546 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=64 time=0.537 ms
64 bytes from 192.168.10.10: icmp_seq=4 ttl=64 time=0.518 ms

--- 192.168.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3077ms
rtt min/avg/max/mdev = 0.518/0.549/0.598/0.029 ms

```

Figure 11. Client-1-VM DHCP addressing, DNS configuration, and internal ping tests.

```

yaboy9k@yaboy9k-Standard-PC-i440FX-PIIX-1996:~$ curl http://www.office.local
<!DOCTYPE html>
<html>
<head>
  <title>CSCE 3530 Network Project</title>
</head>
<body>
  <h1>CSCE 3530 Small Office Network</h1>
  <p>This page is hosted on the internal Server-VM at 192.168.10.10.</p>
  <p>Clients can access this page using the DNS name www.office.local.</p>
</body>
</html>~
yaboy9k@yaboy9k-Standard-PC-i440FX-PIIX-1996:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  _gateway (192.168.10.1)  0.509 ms  0.492 ms  0.470 ms
 2  192.168.4.1 (192.168.4.1)  2.936 ms  2.920 ms  2.901 ms
 3  47.189.58.1 (47.189.58.1)  1.682 ms  1.615 ms  1.541 ms
 4  184.19.247.246 (184.19.247.246)  1.568 ms  1.681 ms  184-19-247-244.2
fb087467977f9ae1c3c3a2f85d79294.ip.frontiernet.net (184.19.247.244)  1.7
49 ms
 5  ae5---0.scr03.dlles.tx.frontiernet.net (74.40.10.210)  2.578 ms ae11-
--0.scr05.dlles.tx.frontiernet.net (74.40.10.208)  3.465 ms  3.362 ms
 6  ae1---0.scr06.dlles.tx.frontiernet.net (74.40.10.110)  2.504 ms ae0--
-0.cbr03.dlles.tx.frontiernet.net (45.52.201.113)  2.587 ms ae1---0.scr06
.dlles.tx.frontiernet.net (74.40.10.110)  2.844 ms
 7  * ae1---0.cbr03.dlles.tx.frontiernet.net (45.52.201.115)  20.680 ms *
 8  * 74-40-26-234.9f4034d1533cc91ba2240b383c8b13a6.ip.frontiernet.net (
74.40.26.234)  3.567 ms *
 9  dns.google (8.8.8.8)  3.527 ms  3.496 ms *
yaboy9k@yaboy9k-Standard-PC-i440FX-PIIX-1996:~$ |

```

Figure 12. Client-1-VM HTTP test and traceroute showing first hop through 192.168.10.1.

Client-2-VM Testing

Client-2-VM also received its IP configuration automatically from DHCP. It received 192.168.10.101/24, used 192.168.10.1 as the gateway, and used 192.168.10.10 for DNS. The same internal, DNS, HTTP, and internet routing tests were successful.

```

Jul 7 16:01
yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996: ~

yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$ ip -br a
lo                UNKNOWN          127.0.0.1/8 ::1/128
ens18             UP                192.168.10.101/24 fe80::be24:11ff:fe58:a194/64
yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$ ip route
default via 192.168.10.1 dev ens18 proto dhcp src 192.168.10.101 metric 100
192.168.10.0/24 dev ens18 proto kernel scope link src 192.168.10.101 metric 100
yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (ens18)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 192.168.10.10
    DNS Servers: 192.168.10.10
    DNS Domain: office.local
    Default Route: yes
yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$ ping -c 4 192.168.10.1
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data.
64 bytes from 192.168.10.1: icmp_seq=1 ttl=64 time=0.309 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=64 time=0.671 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=64 time=0.458 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=64 time=0.451 ms

--- 192.168.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3108ms
rtt min/avg/max/mdev = 0.309/0.472/0.671/0.129 ms
yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$ ping -c 4 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=64 time=0.451 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=64 time=0.559 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=64 time=0.689 ms
64 bytes from 192.168.10.10: icmp_seq=4 ttl=64 time=0.669 ms

--- 192.168.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3069ms
rtt min/avg/max/mdev = 0.451/0.592/0.689/0.095 ms

```

Figure 13. Client-2-VM DHCP addressing, DNS configuration, and internal ping tests.


```

yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=3.18 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=3.86 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=116 time=3.26 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=116 time=3.69 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 3.178/3.496/3.861/0.286 ms
yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$ nslookup www.office.local
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   www.office.local
Address: 192.168.10.10

I
yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$ curl http://www.office.local
<!DOCTYPE html>
<html>
<head>
  <title>CSCE 3530 Network Project</title>
</head>
<body>
  <h1>CSCE 3530 Small Office Network</h1>
  <p>This page is hosted on the internal Server-VM at 192.168.10.10.</p>
  <p>Clients can access this page using the DNS name www.office.local.</p>
</body>
</html>
yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8): 30 hops max, 60 byte packets
 1  gateway (192.168.10.1)  0.580 ms  0.926 ms  0.906 ms
 2  192.168.4.1 (192.168.4.1)  0.828 ms  0.793 ms  0.772 ms
 3  47.189.58.1 (47.189.58.1)  2.913 ms  2.883 ms  2.860 ms
 4  184.19.247.244.2fb087467977f9ae1c3a2f85d79294.ip.frontiernet.net (184.19.247.244)  2.165 ms  184-19-247-246.3b57b0c09c9dc03d046c231ed201e407.ip.frontiernet.net (184.19.247.246)  1.877 ms  184-19-2
47-244.2fb087467977f9ae1c3a2f85d79294.ip.frontiernet.net (184.19.247.244)  2.118 ms
 5  ae11---0.scr05.dl1s.tx.frontiernet.net (74.40.10.208)  2.767 ms  ae5---0.scr03.dl1s.tx.frontiernet.net (74.40.10.210)  2.821 ms  2.853 ms
 6  ae0---0.cbr04.dl1s.tx.frontiernet.net (45.52.201.117)  13.250 ms  ae1---0.scr06.dl1s.tx.frontiernet.net (74.40.10.110)  3.546 ms  ae0---0.cbr04.dl1s.tx.frontiernet.net (45.52.201.117)  10.044 ms
 7  ae1---0.cbr04.dl1s.tx.frontiernet.net (45.52.201.119)  10.024 ms  74-40-26-234.9f4034d1533cc91ba2240b383c8b13a6.ip.frontiernet.net (74.40.26.234)  3.225 ms  ae1---0.cbr03.dl1s.tx.frontiernet.net (45
.52.201.115)  16.630 ms
 8  74-40-26-234.9f4034d1533cc91ba2240b383c8b13a6.ip.frontiernet.net (74.40.26.234)  3.161 ms  3.136 ms *
 9  dns.google (8.8.8.8)  2.889 ms *  2.775 ms
yaboy9k1@yaboy9k1-Standard-PC-i440FX-PIIX-1996:~$

```

Figure 14. Client-2-VM internet ping, DNS lookup, HTTP test, and traceroute.

Verification Results Summary

Test	Result
Router has outside and inside interfaces	Successful
Router can reach the internet	Successful
IPv4 forwarding enabled on router	Successful
nftables NAT/masquerade configured	Successful
Server can ping router and internet	Successful
DHCP server running	Successful
DNS server running	Successful
Apache web server running	Successful
Client-1 received DHCP address	Successful
Client-2 received DHCP address	Successful
Clients resolve www.office.local	Successful
Clients access internal web page	Successful
Clients traceroute through router	Successful

Challenges Faced and Solutions

Challenge	Solution
No DHCP on the internal network at first	The internal vmbr10 network did not have a DHCP server yet, so the router internal interface was configured manually as 192.168.10.1/24. Later, the Server-VM provided DHCP for the client VMs.
Clients needed a DNS server for office.local	BIND9 was installed and configured on the Server-VM. The clients received 192.168.10.10 as their DNS server through DHCP.
Server initially could not resolve www.office.local locally	systemd-resolved was configured to use 192.168.10.10 and the office.local search domain. After restarting systemd-resolved, curl and nslookup worked using the DNS name.
curl and traceroute were missing on client VM	The missing packages were installed with apt, and the HTTP/traceroute tests were then repeated successfully.
Internal clients needed internet access	IPv4 forwarding and nftables NAT were configured on the Router-Firewall-VM so traffic from 192.168.10.0/24 could exit through ens18.

Conclusion

The final network was successfully built and tested in Proxmox. The Router-Firewall-VM provided routing, forwarding, and NAT for the internal network. The Server-VM provided DHCP, DNS, and Apache web services. Both Ubuntu Desktop client VMs received DHCP addresses, used the Server-VM for DNS, accessed the internal web server using `www.office.local`, and reached the internet through the router/firewall VM. The test results confirm that the small office network design is functional and correctly configured.

Submission Checklist

- Final report includes project title, student name, overview, objectives, network design, protocols, configurations, testing, challenges, and conclusion.
- Network topology diagram is included.
- Proxmox VM inventory screenshot is included to document the virtual lab environment.
- Configuration screenshots are included and labeled.
- Testing screenshots are included and labeled.
- Client DHCP leases are verified from the Server-VM.